



**K. K. Wagh Institute of Engineering Education and Research,
Nashik
(Autonomous from Academic Year 2022-23)**

T. Y. B. Tech. Artificial Intelligence and Data Science Pattern 2023 Semester: VI 2301317: Microcontroller and Embedded Systems			
Teaching Scheme:		Credit Scheme:	Examination Scheme:
Theory: 03 hrs/week		03	Continuous Comprehensive Evaluation: 20 Marks InSem Exam: 20 Marks EndSem Exam: 60 Marks
Prerequisite Courses: - 2301206: Digital Electronics and Logic Design			
Course Objectives: • To get familiar with 8051 microcontroller • To understand instruction set and assembly language programming of 8051 • To use C programming to write 8051 programs • To study features of 8051 microcontroller • To get introduced to embedded systems			
Course Outcomes: On completion of the course, students will be able to–			
	Course Outcomes		Bloom's Level
CO1	Explain basics of 8051 microcontroller		2-Understand
CO2	Make use of instruction set to write simple assembly language programs of 8051		3-Apply
CO3	Make use of C to write simple 8051 Programs		3-Apply
CO4	Explain features of 8051 microcontroller		2-Understand
CO5	Illustrate basics of embedded systems		2-Understand
COURSE CONTENTS			
Unit I	Introduction to Microcontroller	(06 hrs)	CO1
Difference between microprocessor and microcontroller, Introduction to the Microcontroller, Features and block diagram of 8051 and explanation, Program Status Word (PSW), Programmers model-register set, register bank, SFRs			
Unit II	8051 Assembly Language Programming and I/O Port Programming	(08 hrs)	CO2
Addressing modes, Introduction to 8051 assembly programming, Structure of assembly language, instruction set: Jump, Loop , Call, arithmetic , logic instructions, 8051 I/O Port Programming			
Unit III	8051 Programming in C	(08 hrs)	CO3
Why program the 8051 in C? , Data types and time delay in 8051 C, I/O Programming in 8051 C, Logic Micro operation in 8051 C, Data Conversion programs in 8051 C			
Unit IV	8051 memory, interrupts and timers/counters	(08 hrs)	CO4
Memory organization on-chip data memory, External data memory and program memory, Memory interfacing-external RAM/ROM interface. CPU timings, Interrupt structure, 8051 Timers/counters, operation modes of 8051 and their programming			
Unit V	Embedded Systems	(06 hrs)	CO5
Introduction to Embedded systems, Characteristics, Challenges, Processors in Embedded systems, Application Domain, Real time systems, Real time task, Hardware Units and devices in an embedded system			

Text Books	
1. Muhammad Ali Mazidi and Janice Gillispie Mazidi, Rolin McKinlay, The 8051 Microcontroller and embedded systems , 2009, Pearson education. 2. V Udayashyankara, M S Mallikarjunaswamy, 8051 Microcontroller, , The McGraw Hill Companies 3. Lyla B. Das, Embedded Systems: An Integrated Approach Pearson , ISBN: 9332511675, 9789332511675 4. Raj Kamal, Embedded Systems: Architecture, programming and Design, 2 nd Edition, McGraw-Hill, ISBN: 13: 9780070151253	
Reference Books	
K. J. Ayala, D. V. Gadre , The 8051 Microcontroller and Embedded systems using Assembly and C., Cengage learning, ISBN 9788131511053	

Strength of CO-PO/PSO Mapping														
	PO												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-
Average	3	-	-	-	-	-	-	-	-	-	-	-	-	-

Guidelines for Continuous Comprehensive Evaluation of Theory Course		
Sr. No.	Components for Continuous Comprehensive Evaluation	Marks Allotted
1	Quiz on Unit 1, Unit 2, Unit 3 each of 15 marks (Total marks will be converted to 15 Marks)	15
2	Theory assignment on Unit 4, Unit 5 each of 10 marks (Total marks will be converted to 5 Marks)	5
Total		20